	Daffodil International University **Department of Computer Science and Engineering (CSE)** **Lab Course Outline**	

Course Code:	CSE 324			
Course Title:	Operating Systems Lab			
Program:	B.Sc. in Computer Science and Engineering			
Faculty:	Faculty of Science and Information Technology (FSIT)			
Semester:	Fall	**Year:**	2024	
Credit:	2.0	**Contact Hour:**	2.5 Hrs/Week	
Course Level:	L3-T2	**Prerequisite:**	CSE 213/231	
Course Category:	Core Engineering			
Instructor Name:	Rabeya Khatun			
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Course Content (from syllabus):
Introduction of the Basic of Linux, its purposes, basic network commands, file management and different types of permission setup, Shell Administration Adding and removing users, Daily administrative works, Configure/installing different software/, Processes in Linux and managing them, System logs and so on, Process Scheduler, Shell script: Introduction to Shell Scripts- Shell programming, Shell program, Decision making and Loop control structure, functions, different programming language in Linux editor environment and implement different Operating system algorithm, scheduling and page replacement algorithms, Deadlock avoidance and detection.

Course Description/Rationale:
The course aims to explore the importance of the operating system, its function and different techniques used by the operating system to achieve its goals as resource manager. The course also explores how application interacts with the operating system and how the operating systems interact with the machine. Also, the course shed light on some of the existing operating systems and how the topics taught in the course are applied in these systems. Some topics in the course are implemented by writing the programs to practically know how.

Course Objective:
1. To learn different types of Linux Commands.
2. To learn shell script.
3. To learn different programming languages in the Linux editor environment and implement different Operating system algorithms.
4. To learn about file management and different types of permission setup.
5. To understand how system processes work and how to manage them.
6. To work with operating systems and analyze them.

Course Outcomes (COs) with Mappings:

After completion of this course, students will be able to:

COs	CO Statements	POs	Learning Domains	Knowledge Profile	Complex Engineering Problem	Complex Engineering Activities
CO 1	**Make** use of the Linux commands and shell programming.	**PO1**	**C3**	**K1 K2**	**EP1**	
CO 2	**Apply** different algorithms of Operating Systems like CPU scheduling algorithm, page replacement algorithms, deadlock avoidance, and detection algorithm with **analyzing** the performance of them.	**PO2**	**C3 C4**	**K3 K4**	**EP2 EP3**	
CO 3	**Able** to design and develop a course project that can have a positive impact on the environment or society.	**PO3 PO9**	**C4 A4 P4**	**K5**	**EP5**	**EA1 EA2 EA4**

Bloom's Taxonomy
Cognitive Domain
C2: Understand
C3: Apply
C4: Analyze
C5: Evaluate
Psychomotor Domain
P2: Manipulating
P3: Precising
P4: Articulating
Affective Domain
A2: Responding,
A3: Valuing,
A4: Organizing

Knowledge Profile
K2: Mathematics
K3: Engineering Fundamentals
K4: Specialist Knowledge
K5: Engineering Design
K6: Technology
K7: Society
K8: Research

CEP Attributes
EP1: Depth of knowledge required.
EP2: Range of conflicting requirements
EP4: Familiarity of issues
EP7: Interdependence

CEA Attributes
EA1: Range of resources
EA2: Level of interaction
EA4: Consequences for society and the environment

Justification of CO-PO Mapping:

- **CO1** covers **PO1** because students need to learn, understand and make use of basic Linux commands to be familiar with Linux operating system and students need to identify processes and file management system problems and build shell programs for those problems..
- **CO2** covers **PO2** because students need to select and apply suitable techniques and algorithms for scheduling, page replacement, deadlock avoidance and detection and analyze the performance of those algorithms.
- **CO3** covers **PO3** because students need to design and develop an operating system project and to demonstrate team work and **PO7** because they need to do teamworks and collaborate to complete the project.

Justification of Knowledge Profile (KP) Mapping:

- **CO1** covers **K1 and K2** because students need fundamental knowledge of computer systems and its environment and special knowledge of how operating systems work and also need to learn different Linux commands and basic programming knowledge in order to write shell scripts.
- **CO2** covers **K3 and K4** because students need theoretical knowledge of processes, process scheduling, scheduling algorithms, deadlock detection and avoidance algorithms, page replacement algorithms and also need special knowledge of shell scripting and computer programming.
- **CO3** covers **K5** because students need to design and develop an operating system project.

Justification of Complex Engineering Problem (CEP) Attributes Mapping:

- **CO1** covers the following CEP attributes:
    - **EP1:** Data from various domains may have different formats, structures, quality, and semantics, which may pose conflicting requirements for data preprocessing and normalization.
- **CO2** covers the following CEP attributes:
    - **EP2:** Complex problems may have wide-ranging or conflicting technical, engineering, and other issues that require different operating system algorithms to address them.
    - **EP3:** Complex problems may have no obvious solution and require abstract thinking and originality to implement suitable OS algorithms that can solve them effectively and efficiently.
- **CO3** covers the following CEP attributes:
    - **EP5:** OS applications and solutions may have wide-ranging or conflicting social implications, such as accountability, authenticity, security, etc., that require careful evaluation and trade-offs.

Justification of Complex Engineering Activities (CEA) Attributes Mapping:

- **CO3** covers the following CEA attributes:
    - **EA1:** OS projects often involve the use of diverse resources, such as powerful computing systems, specialized software tools, and human expertise. Communicating the results and insights of such projects requires the ability to select and utilize the appropriate resources for the intended audience and purpose.
    - **EA2:** OS projects often require the resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues.
    - **EA4:** OS projects have significant consequences for society and the environment in a range of contexts, characterized by their difficulty.

Mapping of Assessment Tools with Course Outcomes (COs):

Assessment Tools	Course Outcomes (COs)	Explanation
Lab Performance	CO1, CO2	This tool measures the student's ability to apply operating system concepts and techniques to prepare shell scripts, algorithms and analyze the performance of those algorithms from various domains. The students are evaluated based on their performance in the lab exercises.
Lab Report	CO1, CO2, CO3	This tool measures the student's ability to evaluate the social, environmental, and economic implications of operating system algorithms and solutions. The students are required to write a comprehensive report and it should also demonstrate the students' collaboration, communication, and management skills.
Lab Project based on CEP	CO3	This tool measures the student's ability to integrate and apply operating system concepts and techniques to solve a real-world problem. The students are required to work in teams to design, implement, and evaluate an operating system solution for a given problem domain. The project should also showcase the students' creativity, innovation, and ethical awareness.

Course Delivery Plan/Lesson Delivery Plan:

Week	Experiment Name and Details	Student Activities	Name of the Tools	Assessment Plan
1	Introduction to Linux, Linux Installation.	Discussion Linux Operating System, features, versions, installation and Ubuntu OS.	Virtual Box, Ubuntu	N/A
2	Course Project discussion and group formation – list of projects, team formation, project plan and deliverables with presentation.	Review different OS projects, team formation and peer discussion on OS projects, Project delivery plan.	Virtual Box, Ubuntu	N/A

3	Introduction to Linux tools- Linux files, Directories, Root, File Permissions, Working with files and directories, Disk related commands.	Learn basic Linux commands, create and remove directories and files and work with those files using those commands.	Virtual Box, Ubuntu	Lab Report and Performance
4	Introduction to Shell Scripts- Shell programming, Shell Variables, Shell Keywords, Write simple Shell program.	Learn shell programming, variables, arithmetic operations and write shell scripts using simple exercises.	Virtual Box, Ubuntu	Lab Performance
5	Decision making and Loop control structure and Functions using shell programming.	Execute shell programs and practice decision making problems and loop control structure. Write functions to solve particular problems of the OS using shell programming.	Virtual Box, Ubuntu	Lab Report and Lab Performance
6	FCFS and SJF scheduling algorithm.	Learn CPU scheduling algorithms, Preemptive and Non preemptive algorithms. FCFS and SJF Algorithms. Analyze and discuss case studies and scenarios.	Virtual Box, Ubuntu	Lab Report and Lab Performance
7	Mid Semester evaluation and feedback.	Review the concepts and techniques learned in the first half of the course. Receive feedback from the instructor and peers.	Virtual Box, Ubuntu	Lab Performance
8	Priority scheduling and Round robin scheduling.	Learn Priority Scheduling Algorithm and Round Robin Scheduling algorithm. Analyze and discuss case studies and scenarios.	Virtual Box, Ubuntu	Lab Performance
9	Deadlock Detection and avoidance algorithms.	Learn Deadlock detection and avoidance algorithms. Implement Bankers algorithm.	Virtual Box, Ubuntu	Lab Report and Lab Performance
10	Page Replacement Algorithms.	Learn FIFO and LRU page replacement algorithms. Analyze and discuss their performance in Page Replacement.	Virtual Box, Ubuntu	Lab Report and Lab Performance

11	Optimal Page Replacement Algorithms.	Learn Optimal page replace algorithm. Analyze and discuss case studies and scenarios.	Virtual Box, Ubuntu	Lab Report and Lab Performance
12	Project proposal and planning.	Form teams and select a project topic related to the operating system. Write a project proposal and plan.	Virtual Box, Ubuntu	Lab Project based on CEP
13	Project implementation and testing.	Implement the project using appropriate OS algorithms, techniques and tools. Test and debug the project.	Virtual Box, Ubuntu	Lab Project based on CEP
14	Project evaluation and improvement.	Evaluate the project using appropriate metrics and methods. Improve the project based on the evaluation results.	Virtual Box, Ubuntu	Lab Project based on CEP
15	Project presentation and demonstration.	Present and demonstrate the project to the instructor and peers. Receive feedback and suggestions.	Virtual Box, Ubuntu	Lab Project based on CEP
16	Project report and documentation.	Write a project report and documentation that describes the project objectives, methods, results, and conclusions.	Virtual Box, Ubuntu	Lab Project based on CEP, Lab Report

Assessment Pattern:

Assessment Tools	**COs**			**Marks (Total=100)**
	CO1	**CO2**	**CO3**	
Attendance	Not Applicable			10
Lab Performance	7	18		25
Lab Report	6	9	10	25
Lab Project based on CEP			40	40
Total Marks	13	27	50	100

Laboratory Experiments and Assessment schemes for Lab performances (25 Marks) and Lab Project based on CEP (40 Marks):

Experiment Name and Details	Teaching Learning Method	CO	Marks of Cognitive Learning Levels				Marks of Psychomotor Learning Levels			Marks of Affective Learning Levels			CO Mark
			C2	C3	C4	C5	P2	P3	P4	A2	A3	A4	
Basic Linux Commands	Lecture, Demonstration, Practice	CO1		2									2
Shell script-variables, keywords arithmetic operations, decision making, loop, control structure and functions	Lecture, Demonstration, Practice	CO1		5									5
FCFS and SJF scheduling algorithm	Lecture, Demonstration, Practice	CO2		2	2								4
Priority scheduling and Round Robin scheduling algorithm	Lecture, Demonstration, Practice	CO2		2	2								4
Bankers algorithm	Lecture, Demonstration, Practice	CO2		2	2								4
FIFO Page Replacement algorithm	Lecture, Demonstration, Practice	CO2		1	1								2
LRU Page replacement algorithm	Lecture, Demonstration, Practice	CO2		1	1								2
Optimal Page Replacement Algorithm	Lecture, Demonstration, Practice	CO2		1	1								2
Project proposal and planning	Group Work, Presentation,	CO3			2				2			1	5
Project implementation and execution	Group Work, Presentation	CO3			5				6			4	15

Project evaluation and improvement	Group Work, Presentation	CO3			5				6			4	15
Project presentation and demonstration	Group Work, Presentation	CO3			1				2			2	5
Total				16	22				16			11	65

Lab Report Writing Assessment Scheme:

Weeks	**Student Activities**	**CO**	**Marks of Cognitive Learning Levels**				**Marks of Psychomotor Learning Levels**			**Marks of Affective Learning Levels**			**CO Mark**
			C2	**C3**	**C4**	**C5**	**P2**	**P3**	**P4**	**A2**	**A3**	**A4**	
Week 3	Write the report using pen, paper and submit	CO1		3									3
Week 5	Write the report using pen, paper and submit	CO1		3									3
Week 6	Write the report using pen, paper and submit	CO2		2	1								3
Week 9	Write the report using pen, paper and submit	CO2		2	1								3
Week 10	Write the report using pen, paper and submit	CO2		2	1								3
Week 16	Write the report using pen,	CO3			5				3			2	10

	paper and submit												
Total				12	8				3			2	25

Learning Materials:

Textbook/Recommended Readings:

Textbooks

1. Unix Shell Programming- Yashavant P. Kanetkar

Reference Books/Supplementary Readings:

1. Teach Yourself Unix Shell Programming in 14 days-Kamran Husain